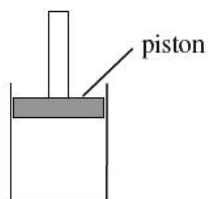


PP Q5

- *5. A fixed mass of an ideal gas is contained in a cylinder fitted with a frictionless piston as shown in the figure below. If the gas is cooled under constant pressure,



- (1) the average separation of the gas molecules will decrease.
- (2) the r.m.s. speed of the gas molecules will decrease.
- (3) the number of collisions per second of the gas molecules on the piston will decrease.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2013 Q3

- *3. In which of the following situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase ?

- (1) The gas is heated under constant volume.
- (2) The gas expands under constant pressure.
- (3) The gas is compressed under constant temperature.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

2016 Q3

- *3. When an ideal gas is heated from 25 °C to 50 °C, the average kinetic energy of the gas molecules will

- A. double.
- B. increase by 41 %.
- C. increase by 8.4%.
- D. increase by 4.1%.

2021 Q3

- *3. For an ideal gas, kinetic theory deduces that $pV = \frac{1}{3} Nmc^2$. Which physical quantity below can be represented by $\frac{3p}{c^2}$?

- A. the total mass of the gas
- B. the volume of one mole of the gas
- C. the density of the gas
- D. the number of gas molecules per unit volume

Book 1 section 4.3: DSE multiple choice

2024 Q3

- *3. The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm^3 of such 'vacuum' at room temperature.

- A. 10^4
- B. 10^6
- C. 10^8
- D. 10^{12}

2025 Q3

- *3. Identical closed vessels X and Y contain carbon dioxide (CO_2) and oxygen (O_2) respectively.



Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same ?

- (1) root-mean-square speed
 - (2) average kinetic energy
 - (3) frequency of collision with the walls of the vessel
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only